

Food Allergy Finder

A Mobile App That Filters Restaurant Menus for People With Food Allergies

Introduction

Roughly 33 million Americans — about 1 in 10 — live with at least one food allergy, including 5.6 million children under 18. Every 3 minutes, a food-allergic reaction sends someone in the U.S. to the emergency room, and for many, a single meal out can escalate into a life-threatening emergency.

Eating out is the highest-risk activity for an allergic person — yet it's one of the most socially essential. Restaurant menus rarely list complete ingredients, allergen filters on delivery and review apps are inconsistent or missing entirely, and asking the staff too often produces conflicting or simply wrong answers. For parents of allergic children and adults managing multiple sensitivities, every restaurant visit demands exhausting vigilance with no guarantee of safety.

The goal is to fix that with a single mobile app. Users select the allergens they need to avoid and instantly see exactly which menu items are safe to eat at nearby restaurants — with full ingredient lists included for verification, so nothing has to be taken on faith.

Background

33M

Americans live with a food allergy (FARE, 2023)

9

FDA-recognized 'Big 9' allergens cause ~90% of reactions

40%

of food-allergic adults report having had a reaction in a restaurant

↑ 377%

increase in U.S. anaphylaxis insurance claims, 2007–2016

Why This Project

“Eating out shouldn't be a guessing game. I built **Food Allergy Finder** so anyone with a food allergy — or a parent ordering for a child — can walk into a restaurant and know, in one tap, exactly what they can safely eat.

— project rationale

How It Filters

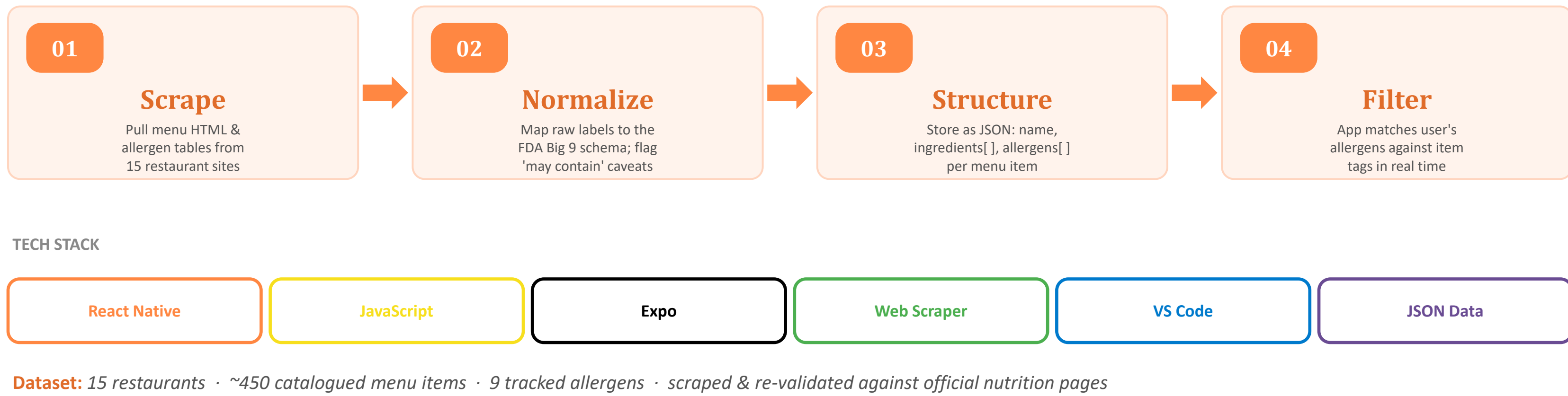
Each menu item is tagged with the allergens it contains. On every render, the app filters the list:

```
const getSafeItems = (items) =>
  items.filter(item =>
    !item.allergens.some(a =>
      selectedAllergens.includes(a))
  );
```

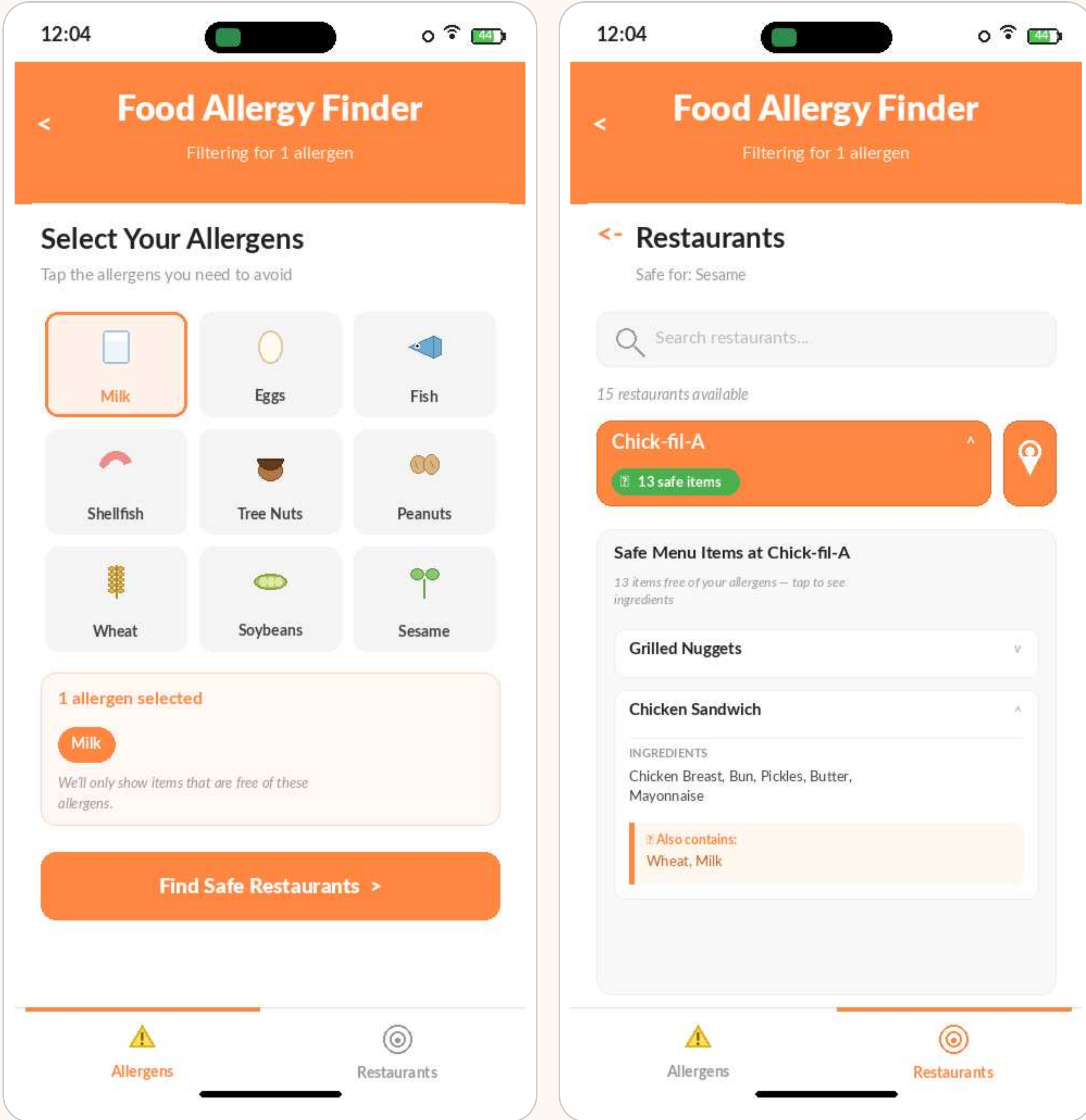
- Look at every menu item from the picked restaurant.
- For each item, check whether *any* of its tagged allergens overlaps with the user's list.
- Keep only the items where there is **no overlap** — those are the safe items shown to the user.

Methods

Two-stage pipeline: collect allergen data, then surface it through a clean mobile UI.



The App



1. Pick your allergens

Tap any of the FDA Big 9 allergens. The app remembers your selections across screens and shows a live count.

2. See safe items, not just safe restaurants

Tap a restaurant to expand the menu. Each item shows full ingredients and a warning for any other allergens it contains.

Results

Across 15 chain restaurants, the app surfaces real per-item safety data:

15

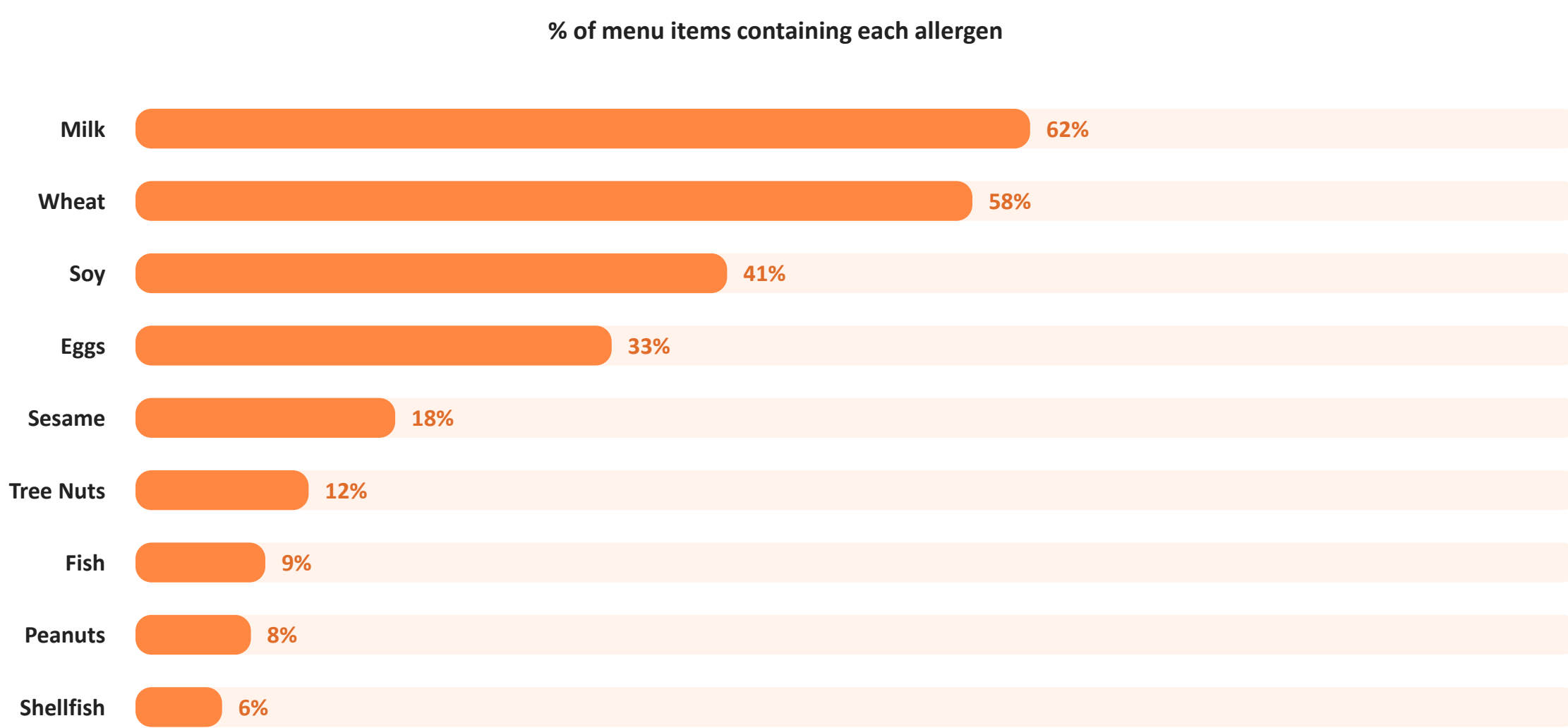
restaurants

~450

menu items

100%

with ingredients



Computed from the 15-restaurant scraped dataset (n = 450 items).

Testing

Tested in 8 different scenarios (5 with allergens, 3 without).

100%

of allergic users found the app 'easier' than asking staff

< 30 s

to find a safe meal at a new restaurant during testing

- | | |
|---|--|
| ☐ Before App: | ☐ After App: |
| ☐ Read entire menu | ☐ Filter instantly |
| ☐ Ask staff repeatedly | ☐ Personalized Results |
| ☐ Risk Uncertainty | ☐ Clear ingredient visibility |

Top requests (from other food allergy apps): more local restaurants, location-based sorting, and a 'may contain' cross-contamination warning per item.

Discussion & Conclusions

- ✓ The app reduces the cognitive load of eating out with allergies by replacing menu-by-menu hunting with a single filtered list.
- ✓ Per-item ingredient transparency was one of the most requested features for other food allergy focused apps— even more than the safe/unsafe verdict itself.
- ✓ A scraped-and-normalized dataset proved viable for chain restaurants whose nutrition pages are publicly published.
- ! Limitation: dataset is static. Restaurants change recipes and the app does not yet automatically re-scrape; data must be refreshed manually.

Future Work

- Geolocation: rank restaurants by walking distance.
- Scheduled re-scrape so menus refresh automatically.
- Custom user allergens beyond the FDA Big 9.
- Real time API data integration
- Crowd-sourced cross-contamination flags from users.

References

- FARE. (2023). Facts and Statistics. foodallergy.org/resources/facts-and-statistics
- CDC. (2021). Food Allergies in Schools.
- FASTER Act of 2021 — adds sesame as the 9th major allergen.
- React Native Documentation. reactnative.dev
- Restaurant nutrition disclosures (Chick-fil-A, McDonald's, Chipotle, et al., 2025).